

**1- The cotton mills born of the Industrial Revolution were staffed with the workers of the former cottage industries, who were no longer required to have ---- in traditional skills but instead simply be capable of operating the machines that now did everything for them.**

- A) extraction
- B) contradiction
- C) proficiency
- D) reluctance
- E) severity

**2- The advent of telecommunications technologies has brought about a remarkable ---- in the transfer of scientific knowledge across the entire world.**

- A) acceleration
- B) perplexity
- C) assumption
- D) exhaustion
- E) fragility

**3- The development and deployment of advanced artificial intelligence models are dominated by a few tech giants and well-funded institutions due to their ---- costs of training and running.**

- A) feasible
- B) excessive
- C) admissible
- D) attainable
- E) reliable

**4- Animals do not dream when they hibernate because hibernation is different from sleep, which is a more physiologically 'active' state, whereas hibernation requires animals to ---- reduce all activities in order to conserve energy.**

- A) coincidentally
- B) substantially
- C) adversely
- D) suspiciously
- E) competitively

**5- By favouring specific variants and carefully choosing which individuals are allowed to breed, farmers have been able to ---- beneficial traits of various plants while eliminating weaker ones.**

- A) amplify
- B) deplete
- C) relieve
- D) constrain
- E) overcome

**6- Microbiologists are often associated with determining the microbes that cause disease, but their work ---- every other facet of life, such as genetics and physiology.**

- A) wards off
- B) strikes against
- C) extends into
- D) roots out
- E) holds back

**7- The US Congress has passed a law requiring that any asbestos-bearing material that is visibly deteriorating ---- by a different material; as a result, billions of dollars ---- on asbestos removal.**

- A) could be replaced / had been spent
- B) must be replaced / are being spent
- C) would have been replaced / will be spent
- D) should be replaced / were being spent
- E) may have been replaced / have to be spent

**8- Birds generally flock together and develop coordination ---- simultaneously, but ---- to use their acrobatic abilities for navigating natural or human-made obstacles.**

- A) acting/would split off
- B) to act/can split off
- C) to be acting / must split off
- D) having acted / may split off
- E) acted/could split off

**9- The light-year, which is the distance that light travels ---- one year, is a convenient unit of measurement to use when discussing distances to the stars in the Milky Way galaxy and ---- the observable universe.**

- A) on / with
- B) from / about
- C) in/throughout
- D) into / over
- E) by / for

**10- Spider monkeys, medium-sized monkeys with very long tails and limbs, usually live in trees and they are very skilled ---- moving around in trees with the help of their long tails which are well-adapted ----holding on to objects.**

- A) off/along
- B) over / about
- C) on /from
- D) at/for
- E) through/towards

**11- Nanotechnology is the field of science and engineering ---- a scale of one to a hundred nanometres, which is ---- the limit of the world that is visible to the naked eye.**

- A) to / along
- B) at/beyond
- C) over / for
- D) about/through
- E) off / from

**12- Many environmentalists believe spending large sums on captive breeding is a waste of resources ---- there is no natural habitat left to release the animals like pandas that bred in captivity.**

- A) until
- B) although
- C) whereas
- D) so that
- E) as

**13- In contemporary usage, the word abacus refers to a computational device with beads sliding on fixed rods, often associated with the Japanese or Chinese; ----, the word abacus has Latin roots, suggesting a rich history in Western as well as Eastern cultures.**

- A) hence
- B) in other words
- C) as a result
- D) however
- E) likewise

**14- Biotin, a member of the water-soluble B-complex group of vitamins, cannot be synthesised by humans; ----, people must obtain this essential micronutrient from dietary sources.**

- A) by comparison
- B) therefore
- C) similarly
- D) for instance
- E) otherwise

**15- ---- great improvements have been made in estimating mean climatic conditions over broad areas, even more could be done to evaluate climatic variability.**

- A) Although
- B) Unless
- C) By the time
- D) As if
- E) When

**16- Normal regulation of the cell cycle ensures the passage of genetic material without mutations and aberrations, ---- proper completion of each phase is critical to the initiation of the following phases.**

- A) rather
- B) instead
- C) so
- D) otherwise
- E) but

**17- In the future, synthetic cells may become factories that deliver drugs to treat human disease, but they are now being intensively investigated ---- answering fundamental questions about biology.**

- A) despite
- B) with the goal of
- C) irrespective of
- D) in comparison to
- E) similar to

**18- Over decades, the numbers of eastern black rhino have declined ---- relentless hunting and poaching, to the extent that there could be as few as 850 left in the wild.**

- A) due to
- B) except for
- C) contrary to
- D) apart from
- E) such as

**19- The potent cocktail of hormones and other proteins that the muscles produce when people exercise is thought to be ---- influential on mood and behaviour ---- they have been nicknamed the 'hope chemicals'.**

- A) either / or
- B) the more / the more
- C) so / that
- D) such / that
- E) whether/or

**20- Animals have varying degrees of natural intelligence, ---- allows them to learn specific kinds of behaviour, either from their own species or from good or bad experiences.**

- A) how
- B) where
- C) whose
- D) why
- E) which

Migrating birds use celestial cues to navigate, much as sailors of the past used the Sun and stars to guide them. **(21)----**, unlike humans, birds also detect the magnetic field generated by Earth's molten core and use it to determine their position and direction. **(22)----** more than 50 years of research into magnetoreception in birds, scientists have been unable to work out exactly how they use this information to stay on course. Recently a group of experts have gained insights into this **(23)----** mystery. New experimental evidence suggests something extraordinary: a bird's compass relies on subtle, fundamentally quantum effects in short-lived molecular fragments, known as radical pairs, **(24)----** photochemically in its eyes. That is, the creatures appear to be able to 'see' Earth's magnetic field lines and, **(25)----** that information, they chart a course between their breeding and wintering grounds.

**21-**

- A) Yet
- B) Thus
- C) Accordingly
- D) For example
- E) Otherwise

**22-**

- A) By means of
- B) Regarding
- C) Despite
- D) Due to
- E) Besides

**23-**

- A) enduring
- B) accurate
- C) eligible
- D) coincidental
- E) vulnerable

**24-**

- A) to be formed
- B) to form
- C) to have been formed
- D) having formed
- E) formed

**25-**

- A) beyond
- B) at
- C) into
- D) with
- E) towards

Scottish microbiologist Alexander Fleming discovered in 1928 that a fungus called *Penicillium notatum* produced penicillin, which became the world's first widely effective antibiotic. *P. notatum* (26)---- high quantities of the drug; however, large-scale production of penicillin became especially crucial when World War II broke out. (27)----, scientists at the University of Oxford sought help from the US Department of Agriculture's Northern Regional Research Laboratory, in Peoria. Andrew Moyer, a microbiologist, (28)---- the task of solving this problem. Moyer's fellow researcher Mary Hunt found a mouldy melon at a Peoria market and brought it to the lab to carefully analyse it (29)---- sophisticated instruments. (30)---- many women conducting research in that area, who were also disregarded, Hunt's contribution to the discovery and study of that mould - which turned out to be *Penicillium rubens* - was undervalued at the time. Moyer's 1944 publication on *P. rubens* mentions Hunt only in the paper's acknowledgements, and the press referred to her as 'Mouldy Mary'.

**26-**

- A) must not generate
- B) cannot generate
- C) may not have generated
- D) could not generate
- E) would not have generated

**27-**

- A) For example
- B) As a result
- C) In brief
- D) Furthermore
- E) In contrast

**28-**

- A) called off
- B) took on
- C) diverted from
- D) interfered with
- E) set back

**29-**

- A) about
- B) around
- C) at
- D) towards
- E) through

**30-**

- A) Rather than
- B) Owing to
- C) Similar to
- D) Regardless of
- E) In favour of

**31- ----; for instance, island-dwelling insects are commonly wingless due to flight increasing the likelihood of being carried out to sea.**

- A) The consequences of flightlessness in insect species can be seriously detrimental
- B) Many wingless forms of insects can be attributed to environmental factors that promote flightlessness
- C) Generalisations in insect populations are difficult to make, due to many structural variations
- D) In winged insects, the relationship between different body parts involved in flight can be complicated
- E) Certain modifications which help some bugs locate their preys have been lost in time

**32- The gut and brain are more intimately connected than any other body systems ----.**

- A) although the electrical stimulation of different central nervous system regions provides convincing evidence of its role in controlling digestive function
- B) before the nerve pathways linking the brain and the gut can be precisely defined by using tracers of nerve cells
- C) because the brain sends signals to the body about almost everything and many of those signals travel to the digestive tract
- D) while the information exchange between the brain and the gut can influence pain sensitivity
- E) given that consuming fermented foods has proven to support the digestive system as well as benefitting the gut-brain axis

**33- ----; therefore, conservation management should be directed at whole landscapes instead of focusing on a single area in order to ensure their survival.**

- A) The activity of modern humans has been one of the most significant factors affecting the landscapes
- B) Ecosystems may be classified according to the dominant type of habitat
- C) Certain animals and plants, including endangered species such as jaguars, range widely across several different ecosystems
- D) Temperate climate ecosystems tend to be simpler than others in terms of numbers of species
- E) The relationship between different organisms in an ecosystem shows the characteristics of the region

**34- ----, the real advances in railroad travel came in the United States, where greater distances fostered a different philosophy.**

- A) Although railroading on the British model, which had short lines, spread throughout Europe
- B) Given that European engineers believed that locomotives could pull heavy loads up hills
- C) Whereas American railroad lines travelled over high mountains using switchbacks
- D) By the time Americans realised that trains could carry people and goods faster than ships
- E) Once British locomotives travelled smoothly on well-constructed railroads

**35- Even if the warming climate were to result in greater moisture and hence more precipitation in coastal zones, ----.**

- A) certain organisms are well-adapted to tolerate drought under changing climate conditions
- B) semi-arid ecosystems have historically been important as livestock producing areas
- C) the change in the Earth's radiation balance affects the rate of precipitation
- D) other areas, such as continental interiors, would likely experience warming without significant additions of precipitation
- E) intensified land use caused by growing human populations is the main driver of desertification

**36- ----; however, these creatures are true minorities, accounting for just a tiny fraction of the animal world as over 97 per cent of animal species are invertebrates.**

- A) Human perception of the animal kingdom tends to focus on relatively large vertebrates such as horses, birds, and fishes
- B) The small and boneless creatures called invertebrates dominate the animal kingdom in terms of population
- C) The members of vertebrate animals exhibit a trend towards increasing size
- D) Most animals are far more diverse and numerous than vertebrates thanks to their high reproduction rates
- E) Vertebrates are difficult to define due to the diversity and complexity of all creatures in the animal kingdom

**37- It should be taken very seriously that 'biocides' are poisons designed to kill living cells; ----.**

- A) on the contrary, biocide concentrates should not be allowed to enter surface water or the environment in general
- B) otherwise, respiratory protection should be utilised in the event of exposure to a biocide through inhalation
- C) instead, an emergency action plan should be in place to deal with accidental spills of biocides
- D) therefore, direct contact with concentrated biocides should be avoided
- E) however, disposal of biocides should be done according to regulations through licensed disposal contractors

**38- ----, it is becoming increasingly challenging to manage the sustainability of coastal ecosystems and their resources without considering both societal and ecological elements.**

- A) Although low-lying coastal areas and small island states are particularly vulnerable to sea level change triggered by climate change
- B) Once the accumulation of coastal garbage poses a threat to environmental health because of microplastics
- C) As long as there are largely-funded programmes aiming to eliminate the environmental hazards along the coastlines in industrialised nations
- D) As human activities in coastal areas have accelerated due to rapidly escalating human population growth
- E) While some coastal species are among the most endangered in the world and have a poorer recovery prognosis than their terrestrial counterparts



**39- ----; by contrast, geography presents the big picture, one that has environmental events and human actions intertwined.**

A) Geology is an important but narrowly focused science that deals with and answers questions about planet Earth

B) Scientists in the field of geology can predict what lies beneath the Earth's surface thanks to precise measurements of its gravitational field

C) Geography draws on a variety of branches of science, including physics and geology

D) The similarities between the limits of geology and geography make it difficult to differentiate between the two fields

E) Geology is about more than just landforms as it is also about patterns, processes, and materials that make up the Earth

**40- Because irrigation accounts for the majority of the water used by humans worldwide, ----.**

A) subsurface irrigation is an emerging technology with high water-utilisation efficiency used by many farms

B) water consumption varies greatly among regions due to differences in economic development

C) achieving a better efficiency of agricultural use is crucial for advancing water conservation

D) technologically advanced irrigation systems incorporate advanced climate-based controls

E) different irrigation systems have been successfully used on fruit trees and certain row-crops

**41- The leading scientific hypothesis has been that insects use the moon's light to steer at night and mistake artificial lights for the moon; ----.**

A) by comparison, insects often invert themselves and try to fly upside down, causing them to plummet

B) on the contrary, the prevalence of light pollution in towns may severely impact their hunting abilities

C) for example, insects will fly in a spiral towards lights, which is not what they do

D) rather, there are circumstances in which the light response might result in unstable flight

E) however, this celestial navigation idea does not explain why insects that only fly during the day also gather around lights

**42- Competition is the conflict between different organisms for control of food, natural resources, and territories, and it can occur between individuals of the same species or between individuals of different species.**

A) Farklı organizmalar arasındaki gıda, doğal kaynaklar ve bölgeler üzerindeki hâkimiyet çatışması olan rekabet, hem aynı türün bireyleri arasında hem de farklı türlerin bireyleri arasında meydana gelebilir.

B) Aynı türün bireyleri arasında veya farklı türlerin bireyleri arasında meydana gelebilen rekabet, farklı organizmalar arasındaki gıda, doğal kaynaklar ve bölgeler üzerindeki hâkimiyet çatışmasıdır.

C) Farklı organizmaların; gıda, doğal kaynaklar ve bölgeler üzerinde hâkimiyet kurma çatışması rekabet olarak adlandırılır ve aynı türün bireyleri arasında olduğu gibi farklı türlerin bireyleri arasında da meydana gelebilir.

D) Rekabet; farklı organizmalar arasındaki gıda, doğal kaynaklar ve bölgeler üzerindeki hâkimiyet çatışmasıdır ve aynı türün bireyleri arasında veya farklı türlerin bireyleri arasında meydana gelebilir.

E) Rekabet; farklı organizmalar arasındaki gıda, doğal kaynaklar ve bölgeler üzerindeki hâkimiyet çatışmasına verilen addır ve aynı türün bireyleri arasında olduğu gibi farklı türlerin bireyleri arasında da meydana gelebilir.

**43- Coral reefs are among the most diverse and productive biological communities in the world; however, their future is under threat due to widespread changes in tropical marine ecosystems.**

A) Mercan resifleri dünyadaki en çeşitli ve üretken biyolojik topluluklar arasında olsa da tropikal deniz ekosistemlerindeki geniş çaplı değişiklikler nedeniyle gelecekleri tehdit altındadır.

B) Dünyadaki en çeşitli ve üretken biyolojik topluluklar arasında yer alan mercan resiflerinin geleceği, tropikal deniz ekosistemlerindeki geniş çaplı değişiklikler sebebiyle tehdit altındadır.

C) Mercan resifleri dünyadaki en çeşitli ve üretken biyolojik topluluklar arasındadır ancak gelecekleri, tropikal deniz ekosistemlerindeki geniş çaplı değişikliklerden dolayı tehdit altındadır.

D) Mercan resifleri dünyadaki en çeşitli ve üretken biyolojik topluluklar arasındadır ancak tropikal deniz ekosistemlerindeki geniş çaplı değişiklikler geleceklerini tehdit etmektedir.

E) Tropikal deniz ekosistemlerindeki geniş çaplı değişiklikler, dünyadaki en çeşitli ve üretken biyolojik topluluklar arasında yer alan mercan resiflerinin geleceğini tehdit etmektedir.

**44- In most rural regions, there are different patterns of settlement that are typically determined by climate conditions and territorial characteristics, and agricultural crops and animals being farmed change from region to region based on these settlement patterns.**

A) oęu kırsal bölgede genellikle iklim koşulları ve toprak özellikleri tarafından belirlenen farklı yerleşim biçimleri bulunur ve yetiştirilen tarım ürünleri ve hayvanlar bu yerleşim biçimlerine bağlı olarak bölgeden bölgeye değişmektedir.

B) oęu kırsal bölgedeki farklı yerleşim biçimleri genellikle iklim koşulları ve toprak özellikleri tarafından belirlenir ve buna bağlı olarak yetiştirilen tarım ürünleri ve hayvanlar arasında bölgeden bölgeye değişiklikler vardır.

C) oęu kırsal bölgedeki farklı yerleşim biçimlerini genellikle iklim koşulları ve toprak özellikleri belirler ve bu yerleşim biçimlerine bağlı olarak yetiştirilip bölgeden bölgeye değişen tarım ürünleri ve hayvanlar bulunmaktadır.

D) oęu kırsal bölgede genellikle iklim koşulları ve toprak özelliklerinin belirledięi farklı yerleşim biçimleri vardır ve yetiştirilen tarım ürünleri ve hayvanların bölgeden bölgeye değişmesi bu yerleşim biçimlerine bağlıdır.

E) oęu kırsal bölgedeki farklı yerleşim biçimlerini belirleyen iklim koşulları ve toprak özellikleri, bu yerleşim biçimlerinde genellikle yetiştirilen tarım ürünlerini ve hayvanları da bölgeden bölgeye değiştirmektedir.

**45- Most electricity is generated at sites such as power stations, wind farms, or nuclear power plants before being transmitted to homes and industries through a complex network known as a power grid.**

A) Elektrięin oęu, elektrik şebekesi olarak bilinen karmaşık bir aę yoluyla evlere ve sanayilere iletilmeden önce elektrik santralleri, rüzgâr çiftlikleri veya nükleer enerji santralleri gibi yerlerde üretilir.

B) Elektrik şebekesi olarak bilinen karmaşık bir aę yoluyla evlere ve sanayilere iletilmeden önce elektrięin oęunun üretildięi yerler arasında elektrik santralleri, rüzgar çiftlikleri veya nükleer enerji santralleri bulunur.

C) Elektrik, karmaşık bir aę olarak bilinen elektrik şebekesi yoluyla evlere ve sanayilere iletilmeden önce oęunlukla elektrik santralleri, rüzgâr çiftlikleri veya nükleer enerji santralleri gibi yerlerde üretilir.

D) Elektrik şebekesi olarak bilinen karmaşık bir aę yoluyla evlere ve sanayilere iletilen elektrięin oęu öncelikle elektrik santralleri, rüzgâr çiftlikleri veya nükleer enerji santralleri gibi yerlerde üretilir.

E) Elektrięin oęu önce elektrik santralleri, rüzgar çiftlikleri veya nükleer enerji santralleri gibi yerlerde üretilerek elektrik şebekesi olarak bilinen karmaşık bir aę yoluyla evlere ve sanayilere iletilir.

**46- Although a few scientists had previously suggested that the continents could move, the idea did not gain acceptance until the German geologist Alfred Lothar Wegener developed the continental drift theory in 1912.**

A) 1912'de kıtasal sürüklenme teorisini geliştiren Alman jeolog Alfred Lothar Wegener'den daha önce birkaç bilim insanı kıtaların hareket edebildiğini öne sürmüştür fakat bu fikir kabul görmemiştir.

B) Kıtaların hareket edebildiğini daha önce birkaç bilim insanı öne sürmüş olsa da bu fikir, 1912'de Alman jeolog Alfred Lothar Wegener tarafından geliştirilen kıtasal sürüklenme teorisi kadar kabul görmemiştir.

C) Daha önce birkaç bilim insanı kıtaların hareket edebildiğini öne sürmüş olsa da bu fikir, Alman jeolog Alfred Lothar Wegener 1912'de kıtasal sürüklenme teorisini geliştirene kadar kabul görmemiştir.

D) Alman jeolog Alfred Lothar Wegener 1912'de kıtasal sürüklenme teorisini geliştirmeden önce birkaç bilim insanı kıtaların hareket edebildiğini öne sürmüştür fakat bu fikir kabul görmemiştir.

E) Daha önce birkaç bilim insanı kıtaların hareket edebildiğini öne sürmüştür ancak bu fikir, Alman jeolog Alfred Lothar Wegener'in kıtasal sürüklenme teorisini geliştirdiği 1912'ye kadar kabul görmemiştir.

**47- The Hubble Space Telescope, which has imaged many stars during different phases of their lives, has played a key role in furthering our understanding of stars' life cycles.**

A) Birçok yıldız yaşamlarının farklı evreleri boyunca görüntülediği için Hubble Uzay Teleskobu, yıldızların yaşam döngülerine ilişkin anlayışımızı daha ileri taşımada önemli bir rol oynamıştır.

B) Hubble Uzay Teleskobu, birçok yıldızın yaşamını farklı evreleri boyunca görüntüleyerek yıldızların yaşam döngülerine ilişkin anlayışımızı daha ileri taşımada önemli bir rol oynamıştır.

C) Yıldızların yaşam döngülerine ilişkin anlayışımızı daha ileri taşımada önemli bir rol oynayan Hubble Uzay Teleskobu, birçok yıldız yaşamlarının farklı evreleri boyunca görüntülemiştir.

D) Hubble Uzay Teleskobu, birçok yıldızın yaşamını farklı evreler boyunca görüntülemekle kalmayıp yıldızların yaşam döngüsüne ilişkin anlayışımızı daha ileri taşımada da önemli bir rol oynamıştır.

E) Birçok yıldız yaşamlarının farklı evreleri boyunca görüntüleyen Hubble Uzay Teleskobu, yıldızların yaşam döngülerine ilişkin anlayışımızı daha ileri taşımada önemli bir rol oynamıştır.

**48- Çalışmalar ağızdaki kasların bir şeyleri sessizce okurken dahi aktif olduğunu göstermektedir ve bu, okuduklarımızı çözümlmek için aslında kelimeleri zihnimize seslendirdiğimiz anlamına gelmektedir.**

A) Showing that the muscles in the mouth are still active when reading something silently, studies imply that sounding out the words in our minds is essential to decode what is read.

B) Studies show that the muscles in the mouth are still active when reading something silently, and this implies that we are essentially sounding out the words in our minds to decode what we read.

C) Studies which show that the muscles in the mouth are still active when something is read silently imply that the words are essentially sounded out in our minds so that what we read can be decoded.

D) Studies show that the muscles in the mouth are still active when something is read silently, and this implies that we decode what is read by essentially sounding the words out in our minds.

E) Studies show that the muscles in the mouth are still active when reading something silently, which essentially implies that what we read is decoded when we sound the words out in our minds

**49- Danimarka'daki mühendisler, dondurmayı daha lezzetli, maliyet açısından daha hesaplı ve hatta yüksek sıcaklıklarda çok daha kolay taşınabilir hâle getirmek için suyun, yağın ve havanın kimyasını değiştiriyor.**

A) In order to create ice cream that could be more delicious, more cost-effective, and even easier to transport in high temperatures, engineers in Denmark are modifying the chemistry of water, fat, and air.

B) The chemistry of water, fat, and air is being modified by engineers in Denmark who aim to make ice cream more delicious, more cost-effective, and even easier to transport in high temperatures.

C) Engineers in Denmark are making ice cream more delicious, more cost-effective, and even easier to transport in high temperatures by modifying the chemistry of water, fat, and air.

D) Engineers in Denmark are modifying the chemistry of water, fat, and air to make ice cream more delicious, more cost-effective, and even easier to transport in high temperatures.

E) Engineers in Denmark are modifying the chemistry of water, fat, and air so that they can create ice cream that is more delicious, more cost-effective, and even easier to transport in high temperatures.

**50- Yeryüzündeki orman, mera ve sulak arazilerin çoğu tarımsal ve kentsel arazilere dönüştürülmektedir ve o habitatlarda yaşayan türlerin sayısı, yeni çevreye uyum sağlayamadıkları için büyük ölçüde azalabilir.**

A) It is the transformation of much of the forests, grasslands, and wetlands on earth into agricultural and urban areas that can drop the number of species living in those habitats as they are unable to adapt to the new environment.

B) As much of the forests, grasslands, and wetlands on earth is being transformed into agricultural and urban areas, the number of species living in those habitats can drop greatly due to their inability to adapt to the new environment.

C) Much of the forests, grasslands, and wetlands on earth is being transformed into agricultural and urban areas, and the number of species that live in those habitats can drop greatly because they are unable to adapt to the new environment.

D) Much of the forests, grasslands, and wetlands on earth is being transformed into agricultural and urban areas; moreover, species living in those habitats are unable to adapt to their new environment, which may lead to a great drop in their number.

E) As a result of the transformation of much of the forests, grasslands, and wetlands on earth into agricultural and urban areas, the number of species that live in those habitats and are unable to adapt to the new environment can drop greatly.

**51- Mikroorganizmalar ve özellikle bakteriler, küçük boyutları ve kolayca ayırt edilebilen değişken özelliklerinin olmaması nedeniyle erken dönem genetikçileri tarafından genellikle göz ardı edilmiştir.**

A) Early geneticists generally ignored microorganisms, and particularly bacteria, due to their small size and the lack of easily recognisable variable traits.

B) Microorganisms, and especially bacteria, which lack easily recognisable variable traits due to their small size, were generally ignored by early geneticists.

C) Microorganisms, and particularly bacteria, were generally ignored by early geneticists because of their small size and the lack of easily recognisable variable traits.

D) Early geneticists generally ignored microorganisms, and especially bacteria, since they are small in size and lack easily recognisable variable traits.

E) Microorganisms, and particularly bacteria, were generally ignored by early geneticists, as they are small in size, lacking easily recognisable variable traits.

**52- Eşittir işaretinin matematikte birden fazla anlama sahip olması bilgisayarlar için sorunlara sebep olmaktadır ve bunu düzeltmek matematiğin temellerini yıkmak anlamına gelebilir.**

A) Issues that computers have are caused by the fact that the equal sign has more than one meaning in mathematics, and fixing this may mean the destruction of the foundations of mathematics.

B) The equal sign has more than one meaning in mathematics, which causes issues for computers, and fixing this would mean that the foundations of mathematics could be destroyed.

C) In mathematics, the equal sign has more than one meaning, causing issues for computers, and fixing this means that the foundations of mathematics may have to be destroyed.

D) Causing issues for computers, the equal sign has more than one meaning in mathematics, and fixing this may mean the destruction of the foundations of mathematics.

E) That the equal sign has more than one meaning in mathematics is causing issues for computers, and fixing this may mean destroying the foundations of mathematics.

**53- Hafıza, dil ve dikkati içeren bilişsel becerilerimiz yaşla birlikte doğal olarak körelir ancak son bilimsel araştırmalar, evcil hayvan sahibi olmanın bu süreci yavaşlatabileceğini göstermektedir.**

A) It has been shown by recent scientific studies that having a pet might slow down the process of natural deterioration of our cognitive abilities, including memory, language, and attention, with age.

B) While our cognitive abilities, including memory, language, and attention, naturally deteriorate with age, this process might be slowed down by having a pet, as shown by recent scientific studies.

C) Our cognitive abilities, including memory, language, and attention, naturally deteriorate with age; however, recent scientific studies show that having a pet might slow this process down.

D) As we age, our cognitive abilities, including memory, language, and attention, naturally deteriorate, yet recent scientific studies show that this process might be slowed down by having a pet.

E) Recent scientific studies show that it is natural for our cognitive abilities, including memory, language, and attention, to deteriorate with age, but having a pet might slow this process down.

**54- The International Mathematical Olympiad (IMO), is considered the world's most prestigious competition for young mathematicians. Correctly answering its test questions requires mathematical ability that artificial intelligence (AI) systems generally lack. Yet, a tech company developed a new AI system which was trained on geometry and could answer some IMO geometry questions as well as humans. ---- Thus, the tech company has now released another AI model which can solve a wider range of problems, and a new version of the previous one. When the team tested them on the questions of IMO 2025, they answered four out of six questions correctly, giving them a score of 28 out of a possible 42 points, which was enough to win a silver medal.**

A) The AI model that can solve olympiad grade mathematics problems has triggered other tech companies to produce more sophisticated models.

B) Furthermore, the developer company had to find a solution since the model came up with an answer far later than the maximum time allowed for each question.

C) However, it could not answer questions from other mathematical disciplines, such as number theory and algebra, which is necessary to win an IMO medal.

D) But many AI models rely on a trial-and-error approach called reinforcement learning, where the system finds its own way to solve a problem over many attempts.

E) Many AI researchers around the world suggest that the rate at which AI improves in the field of mathematics is breathtaking

**55- The microclimate in an urban area as compared to that of the countryside is usually quite obvious. A city, for example, is generally characterised by having haze, higher temperatures, lower wind speed, and reduced radiation. The countryside, on the other hand, is characterised by clean, clear air, lower temperatures, and high wind speeds and radiation. ---- Instead of a mixture of soil or M vegetation, the surface layer is covered with concrete, glass, and stone surfaces ranging to heights of several hundred feet. These materials shed and carry away water, absorb heat, and block and channel the passage of winds, thus altering the climate conditions in the area.**

A) These different microclimatic conditions should come as no surprise to anyone when the materials used in building cities and their effects are considered.

B) To most people, these differences in the infrastructural characteristics of the city and countryside are hardly noticeable.

C) When patterns in the weather occur over a long period, they can be used to define the climate of a region, which affects the conditions of a city.

D) The geographical, biological, and man-made features that make local climate different from the general climate is called a microclimate.

E) For example, climates have been classified into zones by vegetation, moisture index, and even measures of human discomfort.



**56- Ocean alkalinity enhancement is a proposed method of removing excess carbon dioxide (CO<sub>2</sub>) from the atmosphere. It involves adding alkaline substances, such as basalt or calcium oxide, to oceanwater to increase its pH. This helps to convert dissolved CO<sub>2</sub> in seawater into carbonates and bicarbonates, unlocking capacity for the oceans to draw down further CO<sub>2</sub> from the atmosphere. ---- However, releasing vast quantities of alkaline substances into the ocean could have catastrophic consequences for the marine food web.**

A) Efforts to boost the ocean's carbon-storing capacity by making seawater more alkaline could have unfavourable outcomes, researchers have warned.

B) In theory, ocean alkalisation could remove many billions of tonnes of CO<sub>2</sub> from the atmosphere, helping to limit the impacts of climate change.

C) Interestingly, shallow coastal waters may respond differently to ocean alkalinity enhancement compared with deep-ocean settings.

D) Naturally occurring minerals such as basalt and olivine were found to be ineffective at boosting ocean carbon absorption because they do not dissolve well in ocean water.

E) In fact, researchers have already investigated how the release of alkaline minerals would affect the cycling of carbon in the ocean by assessing the impact of alkaline mineral release.

**57- Beginning in the 16th century, as botanists explored more areas of the world and identified more and more species, the problem of how to best classify all these species became critical. By the mid-19th century, a number of botanists had devised classification schemes that were based on using a large number of traits. As the century progressed, there was also greater interest not only in plants' external form or morphology, but also in internal structures and in microscopic examination of plant tissue. ---- This increased morphological knowledge improved classification of plants by providing more information on which to base categorising decisions.**

A) Plants thriving in dry sites were observed to have several external features that appear to be adaptations for limiting water loss.

B) The diversity of plant species and even the great number of forms within a single plant made the study of plants a challenging task.

C) The basic classification system used by 19th-century botanists grew out of the work of botanist Carl Linnaeus in the 18th century.

D) With this trend came more work on plant development, on how plant structures arose, enlarged, and changed over time.

E) The study of biogeography deals with the relationships between plant form and geographical characteristics such as climate and terrain.

**58- When John Dalton put forward his atomic theory, chemistry was not yet a quantitative science. Chemists did not know how elements combined or whether they always combined in the same proportions. ---- They thought that metals could be alloyed in whatever proportions one desired and that the quantity of the ingredients of a dye could vary. They did not see why chemical compounds should be any different. Dalton changed all that by proposing that the proportions of atoms comprising a molecule could not vary.**

A) Moreover, many chemists believed that the ratio of different elements in a compound was variable.

B) Dalton revolutionised chemistry, not by reintroducing an old idea but by formulating a workable theory of the formation of chemical compounds.

C) The complexity of the materials with which chemists worked made generalisations about their behaviour difficult.

D) Chemists confined themselves to working with solids and liquids and paid no attention to the gaseous state of matter.

E) Dalton's theory did not lie in the assumption that matter is composed of indestructible atoms, which was a long-established thought.

**59- One common attribute of societies considered modern is the high degree to which they use technology. This statement pertains not only to the domain of industrial manufacturing and the organisation of work, but also to the private everyday world in which each member of society lives. The way people lead their daily lives and take part in society is facilitated or complicated by residential infrastructure, public and private means of transportation, depending on the design, ease of handling and accessibility of all these things. ---- Communication, playing sports, keeping house, enjoying leisure time - all these are no longer conceivable in modern industrialised societies in the absence of technological support.**

A) In the domestic environment, the application of technology reduces physical hardship.

B) Technology is a supportive tool for persons of all ages, but it is particularly important for the elderly.

C) While technological innovations are often welcomed in by a large number of advocates, they may have some opponents as well.

D) There are few domains of contemporary life that would be what they are now without technology.

E) Yet, in some remote parts of the world, there are traditional societies lacking access to technological advancements.

60-

**(I)** The elkhorn coral was once a very common species, but white band disease in the 1980s killed off 97 per cent of the population, and today it is still critically endangered. **(II)** It is not only a keystone species, but as one of the fastest growing corals, it is an ecosystem engineer. **(III)** Climate change contributes to this by causing mass coral bleaching, while rising sea temperatures kill the coral by forcing it to release the algae that it feeds on. **(IV)** It suffers from disease, unsustainable fishing, pollution, and ocean acidification as well. **(V)** Humans also play a role in its demise by standing on reefs and damaging it when snorkelling or diving.

A) I B) II C) III D) IV E) V

61-

**(I)** Everything scientists know about particle physics, summed up in a theory called the Standard Model, suggests that the Big Bang should have created equal quantities of matter and antimatter. **(II)** Physicists have proposed many hypotheses to explain this, and some of these proposals include side effects that scientists can hope to detect. **(III)** According to this theory, when matter and antimatter particles collide, they destroy one another, so the mass created when the universe was born should have been completely wiped out, leaving an empty, featureless cosmos containing only light. **(IV)** That there was enough leftover matter after this great annihilation to form galaxies, stars, planets and even us but almost no antimatter is known as the matter-antimatter imbalance. **(V)** This existential anomaly is one of the great outstanding mysteries of modern physics.

A) I B) II C) III D) IV E) V

62-

**(I)** Volcanic energy cannot be harnessed; that is, controlled and collected, but in a few places heat from the earth, called geothermal energy, can be collected. **(II)** Usually, engineers try to collect this heat in the rare places where the Earth's crust has trapped steam and hot water. **(III)** Here, they drill into the crust and allow the heat to escape, either as steam or as very hot water. **(IV)** Pipes carry the hot water to a plant, where some of the steam is allowed to 'flash' or separate from the water, eventually turning a turbine or generator to make electricity. **(V)** It has a major environmental benefit because it compensates for the air pollution that would have been produced if fossil fuels were the energy source.

A) I B) II C) III D) IV E) V

63-

**(I)** Every technological leap brings with it challenges as well as opportunities. **(II)** The smartphone has transformed the manner in which we communicate, the search engine has changed how we can find information, and social media platforms have revolutionised the way we interact with each other. **(III)** Our mounting reliance on technology, however, has had an adverse impact on our cognitive skills. **(IV)** We cultivate awareness of our own individual rhythms of attention and accordingly structure our daily schedules so that tasks demanding high concentration align with peaks of focus. **(V)** We no longer read in-depth as much because our attention span has become shorter than it used to be as we are frequently distracted by our personal devices.

A) I B) II C) III D) IV E) V

64-

**(I)** Given the complexity of 21st-century machines, it is curious that a phrase published in the 1670s confirms their mechanics. **(II)** The phrase "as the extension, so the force", known as Hooke's law, states that the deformation of an elastic object, such as a metal spring, is proportional to the force applied to it. **(III)** People use Hooke's law to analyse an astonishing variety of materials - metal, concrete, rubber, even bone. **(IV)** The law remains in effect only if the object stays elastic that is, if it returns to its original shape when the force is removed. **(V)** When the force becomes too large, Hooke's law does not apply.

A) I B) II C) III D) IV E) V

65-

**(I)** Dry, dusty soil and blazing hot sun are challenging conditions for plant growth, and species unsuited to such sites will soon start to suffer. **(II)** However, some of the most beautiful plants and flowers have adapted to thrive in these areas, including a whole host of colourful blooms, and sculptural shrubs and trees. **(III)** Unlike sun-loving plants, shade-loving plants must absorb as much of the available light as possible, and as a consequence, their foliage is generally large and thin. **(IV)** One can often tell when plants are suited to hot, dry sites, as many share key characteristics that help them survive. **(V)** These include small leaves to reduce moisture loss, silvery foliage to reflect light, and hairs that trap water droplets, protecting the surface from the sun and reducing evaporation.

A) I B) II C) III D) IV E) V

Recent headlines hinting at the discovery of life on Mars may sound exciting, but the truth is that the evidence found to date is far from conclusive. It would be great to find unambiguous proof, for example microscopic bacteria or ancient fossils, both of which are plentiful on Earth. Instead, what we see on Mars are exciting clues that may or may not have a biological origin. The hope is that these will eventually accumulate to the point where Martian life, probably in the distant past, is the most likely explanation. That is the main motivation behind several current and planned missions to the Red Planet, but there was intense speculation before the advent of space travel. By the 19th century, telescopes were already powerful enough to show Martian surface features that reminded many people of Earth - some of them real, such as seasonally changing ice caps and mountainous regions, as well as several misinterpretations including lakes, canals, and large areas of vegetation. Speculation at that time focused on the idea that the inhabitants of Mars had similar levels of intelligence and civilisation to ourselves. This view was particularly prevalent in science fiction, but it was taken seriously by many scientists as well. It was not until the first spacecraft flew past the Red Planet in the 1960s that the hope of finding anything resembling human life there was shattered. In fact, many possibilities were crossed off the list at that point - everything from grass and trees to dinosaurs or small mammals - owing to the ultra-thin Martian atmosphere, the scarceness of water, and the lack of any visible traces of life.

**66- Which could be understood from the passage?**

- A) Speculations about life on Mars have been steadily increasing in recent years thanks to ambitious explanations of scientists.
- B) The evidence gathered up to now has not been sufficient enough for scientists to make reliable assumptions about the existence of life on Mars.
- C) The clues found on the surface of Mars have supported the fact that the scarceness of water is the main obstacle to biological activity.
- D) Until the 19th century, scientists had hypothesised about the existence of microscopic bacteria rather than intelligent life forms.
- E) After space travel became a reality in the 1960s, science fiction works began to feature the possibility of life on Mars.

**67- Prior to the advent of space travel, there was much speculation about Mars ----.**

- A) given that science fiction writers had been focusing closely on the existence of dinosaurs or small mammals
- B) as scientists thought that the surface features revealed by telescopes could ensure the survival of microscopic bacteria
- C) because sophisticated telescopes confirmed the existence of some Earth-like features
- D) while telescopes greatly helped to generate accurate data on ancient life forms such as bacteria
- E) even though the possibility of extraterrestrial human life had been largely eliminated

**68- According to the passage, the possibility of finding anything comparable to human life - ---.**

A) was dismissed when telescopes discovered that there was no grass or trees on the surface of Mars

B) captured the attention of science fiction writers but did not greatly impact scientists

C) was ruled out after the first space mission proved that Mars' environment is not suitable for it

D) was promoted with the increasing number of studies on the atmospheric density of Mars

E) was the reason behind developing powerful telescopes in the 1960s

Air quality inside buildings is related to a diverse range of chemical, physical, and biological factors. In any situation, the importance of these many influences can vary greatly, depending on the emission rates of various chemicals, the frequency with which inside air is exchanged with outside air, the efficiency of atmospheric circulation within the building, and numerous other factors. In response to the need to conserve energy, modern buildings are well insulated to retain their heat in winter and their coolness in summer. Such buildings receive almost all of their inputs of relatively clean, outside air through their carefully designed ventilation systems. Such systems have only a few, discrete intakes of outdoor air, and outputs of used air back to the outside, as well as particular, internal-circulation characteristics. Of course, the ventilation characteristics of many recently constructed modern buildings have a substantial influence on the quality of the internal atmosphere of the structure. When ventilation systems are operated to save energy, there are relatively few exchanges of indoor air with relatively clean, outdoor air. Sometimes, too much attention to the efficiency of energy use may result in harmful consequences in airtight buildings, such as the build-up of excessive concentrations of indoor air pollutants due to ongoing emissions of chemicals within the building.

**69- According to the passage, maintaining good indoor air quality can be difficult because ----.**

- A) ventilation systems are unable to remove all pollutants successfully
- B) most buildings usually depend on uncontrolled natural air flow
- C) there is a wide range of contributors with varying degrees of significance
- D) insufficient insulation causes pollutants to pile up indoors
- E) the outside air to be used may have already been affected by biochemical factors

**70- It can be understood from the passage that many modern buildings ----.**

- A) are so heavily insulated that natural ventilation is rarely used to improve atmospheric circulation
- B) mainly use mechanical ventilation systems hampering the excessive concentrations of pollutants
- C) reduce indoor pollutants by increasing the frequency and duration of air exchanges
- D) are insulated to conserve energy and therefore their air circulation requires well-built ventilation systems
- E) allow for regular air exchange when toxic chemicals rapidly accumulate within a building

**71- Which could be inferred from the passage?**

- A) Internal ventilation systems in airtight buildings are well-equipped to tackle chemical emissions.
- B) The effects of chemical emissions within buildings change based on the level of insulation.
- C) When outdoor air is used in ventilation systems, the harmful effects of chemicals could be highly increased.
- D) Overemphasis on energy-saving strategies can lead to hazardous circumstances within indoor environments.
- E) The drawbacks of reducing air exchange to conserve energy have led engineers to come up with better ventilation systems.

Noise-cancelling headphones work by using microphones that listen to external sounds. Using clever electronics, these sounds are then 'cancelled' by playing an inverted wavelength to the listener. It is like how a vehicle's active suspension dampens vibrations from a bumpy road. The result for the listener is beautifully clear audio with very little audible background noise. The noise-cancelling headphones can even help protect ears from excessive volume - less background noise means devices do not have to play their sounds so loudly - so parents worldwide have been enthusiastically encouraging their children to wear them. This sounded like a win-win until we started hearing about young people who were increasingly suffering from auditory processing disorder. These individuals often struggle to understand sounds and spoken words with distracting background noise. The cause has been linked to the massive increase in young people using noise-cancelling headphones. Instead of their brains developing normally and learning to filter out the noisy world we live in, they never give their grey matter a chance to wire itself up appropriately. Our brains are like muscles: they develop and adapt in response to their environment. Luckily, auditory therapy can help to rewire the brain, but the best solution is to listen to the world around us a little more before it becomes a problem.

**72- It is stated in the passage that noise-cancelling technology ---.**

- A) is a new kind of technology whose effects on our auditory function are not known just yet
- B) may protect the sense of hearing since it provides clear audio free from background noise
- C) has been harshly criticised by parents since its harmful effects on children's brain development were revealed
- D) is often preferred by drivers since it cancels out the sounds of the vehicle's suspension on a rough road
- E) makes it difficult for children to hear audible sounds even in the absence of background noise

**73- According to the passage, auditory processing disorder ----.**

- A) is widespread among individuals who listen to loud music constantly through headphones without noise-cancelling feature
- B) necessitates the sufferer to wear a sophisticated hearing device which is capable of cancelling background noise
- C) could result from the underdevelopment of the brain's noise-filtering function due to the excessive use of noise-cancelling headphones
- D) paralyses individuals' hearing mechanism in such a way that they are unable to hear background noise anymore
- E) causes the individuals' sense of hearing to become extremely insensitive to all kinds of noises

**74- The author's tone towards noise-cancelling technology is ----.**

- A) admiring
- B) neutral
- C) ironic
- D) critical
- E) encouraging



Lithium-ion batteries are ubiquitous, not just in earbuds, phones, and cars, but also in massive facilities that store renewable energy for when the Sun does not shine or the wind dies down. But lithium itself is relatively scarce and available from just a few countries. A world that runs on renewable energy would need 200 times more battery capacity than exists today - and that probably means a different kind of battery. A decades-old technology may be rising to the challenge: batteries that use sodium rather than lithium ions to carry and store charge. Sodium is everywhere, in seawater and salt mines, so supply and cost are not a problem. But the metal is not as good at storing charge as lithium because its ions are three times bigger, hampering their ability to slip in and out of existing battery electrodes. Labs worldwide are developing new electrode materials to address that shortcoming, and in the past 6 months, several groups have announced sodium batteries that hold as much energy as low-end lithium cells. Commercial sodium-ion batteries are starting to roll off the assembly lines for electric vehicles, scooters, and grid power storage. Researchers caution, though, that sodium batteries are not ready for widespread deployment, but are promising to replace lithium-ion batteries which are no longer feasible in today's green world.

**75- According to the passage, lithium-ion batteries ----.**

- A) are preferred over sodium-ion batteries as the latter is a scarce element found only in some parts of the Earth
- B) are versatile in that they can be found in everything from our personal technological devices to facilities storing renewable energy
- C) should be researched further as a technological back-up to sodium batteries due to the needs of today's energy-bound sectors
- D) are expected to rise to peak in their performance after they become supported by sodium-ion batteries
- E) have been developed recently to aid sodium-ion technology which fails to meet the demand of massive facilities

**76- It can be understood from the passage that sodium-ion batteries ----.**

- A) are not as feasible as lithium-ion batteries, thus diverting the attention of researchers to other materials
- B) have increased in price due their utilisation in place of lithium-ion batteries
- C) can provide 200 times more battery capacity than existing methods thanks to new electrode materials
- D) were at first not comparable with lithium-ion batteries in terms of retaining charge due to their larger ion size
- E) have been enormously in demand for the mass production of electric vehicles and scooters

**77- Which of the following can be concluded from the passage?**

- A) Scientists should focus on improving the efficiency of lithium-ion batteries instead of wasting time and resources on sodium-ion batteries.
- B) Batteries based on sodium could aid a future green energy economy if they can match lithium battery performance.
- C) The economic incentive for a shift from lithium-ion batteries to sodium-ion batteries is lacking for now.
- D) Sodium batteries are struggling to reach even half the energy capacity of the low-end lithium cells.
- E) Initially, sodium batteries could not have been used widely due to their high production costs.

It does not take an expert in environmental science to realise that the world is changing. Is it getting warmer? It depends on where one lives. Global warming does not necessarily mean that local climates will be warmer or suffer serious harm. For most of the people on the planet, the changes in climate are almost imperceptible; however, that is not the case for many plants and animals. Many species are experiencing dramatic changes in their patterns of growth and migration, which are closely linked to the conditions of their environment. Given that different species react to climatic changes in different ways and many species are not resilient to these changes, their populations might shrink. For now, while much of the world is experiencing a moderate level of warming, some plants and animals can overcome it by escaping to higher latitudes or elevations. However, these escape routes are limited. Many species do not have anywhere to go, and therefore they could be dramatically affected.

**78- It is understood from the passage that global warming ----.**

- A) has more devastating impacts on humans than it does on plants and animals
- B) is a source of disagreement between experts and the public as to which regions are warmer
- C) can also be studied in detail by focusing on local climates where temperatures are rising
- D) has led experts to study the negative effects of climatic changes more intensively
- E) is a phenomenon whose effects are not going to be the same everywhere

**79- It is clearly stated in the passage that many species ----.**

- A) are still adversely affected by climatic changes although they have fled to new places
- B) may be threatened with population decline because of their vulnerability to climatic changes
- C) might not be able to protect their numbers despite finding a variety of escape routes
- D) face serious risks as they have not been able to change their growth and migration patterns
- E) have the capacity to adapt to the changes in their environments

**80- What is the primary purpose of the author?**

- A) To describe the characteristics of the places to which most species migrate due to climatic changes
- B) To warn the scientific community about the rapid pace of global warming so that immediate action can be taken
- C) To discuss why many species move to higher latitudes or elevations although environmental conditions are unfavourable there
- D) To exemplify the ways different species have evolved to avoid the harmful effects of global warming
- E) To highlight that species differ from each other in how severely they might be affected by global warming

Cevaplar

1-C

2-A

3-B

4-B

5-A

6-C

7-B

8-B

9-C

10-D

11-B

12-E

13-D

14-B

15-A

16-C

17-B

18-A

19-C

20-E

21-A

22-C

23-A

24-E

25-D

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27-B

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67-C

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73-C

74-D

75-B

76-D

77-B

78-E

79-B

80-E